PulseGuard

Credit-Risk Model Governance Portfolio

Interview Defense Master Pack v5

LightGBM + Isotonic Calibration | AUC=0.7769 | ECE=0.0034

307,511 applicants | 140 features | 7 relational tables | 57.4M rows

GOLD checkpoint 89.3% | All claims artifact-traced

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1. Project Identity & Honest Framing

What PulseGuard Is

PulseGuard is a **credit-risk model governance portfolio project**. It demonstrates the complete ML governance lifecycle on the Home Credit Default Risk public dataset: raw multi-table data ingestion, feature engineering across 7 relational tables, leakage-audited splits, a 12-model baseline tournament, Optuna HPO, isotonic calibration, 9-component composite champion selection, score-band policy, SHAP reason codes, fairness proxy audit, drift/PSI monitoring, and a local BM25+LLM governance assistant.

What PulseGuard Is NOT

- A production lending system or regulatory-compliant credit scoring deployment
- A legally certified adverse action notice generator
- A model approved for underwriting in any jurisdiction
- A system trained on real bank customer data
- A fairness-certified model under ECOA or equivalent regulation

Strongest Safe One-Liner

"End-to-end credit-risk governance stack: tuned LightGBM on 307k Home Credit applicants, calibrated PD scores (ECE=0.0034), score-band policy, SHAP reason codes stable across 30 bootstrap resamples, fairness proxy audit, and a local RAG/LLM governance assistant — all under ASSISTIVE_ONLY, HUMAN_REVIEW_REQUIRED constraints."

Why This Is Governance, Not Just a Model

The model is only one layer. PulseGuard also produces: champion selection rationale (composite score, not AUC alone), calibrated probabilities with ECE documentation, score-band policy with semantic justification, SHAP reason codes with stability evidence, a fairness skeleton, drift monitoring baseline, RAG/LLM assistant with hard ASSISTIVE_ONLY constraints, and a full evidence ledger mapping every claim to a file.

Label Legend

Label	Meaning
[IMPLEMENTED]	Code exists, runs, and produces a verified artifact or measured output
[PARTIALLY VERIFIED]	Code runs but validation is limited by public data, proxy fields, or sample constraints
[SIMULATED]	Deliberately scenario-based or synthetic — e.g. cost matrix, policy demo
[FUTURE]	Not built; explicitly absent or deferred

2. Dataset & Scope

Home Credit Default Risk (PRIMARY) [IMPLEMENTED]

Field	Value
Source	Kaggle — Home Credit Default Risk public competition
Applicants	307,511
Default rate	8.07% (scale_pos_weight = 11.39)
Total rows	57.4M across 7 side-tables
Geography	Single Eastern European portfolio
Timestamps	None — out-of-time split not possible
Split	Stratified random 60/20/20, seed=42, DR preserved across all splits

Tables Used

Table	Rows	Signal Domain
application_train.csv	307,511	Core applicant fields — base table
bureau.csv	1.7M	Credit bureau history per applicant
bureau_balance.csv	27.3M	Monthly bureau balance snapshots
previous_application.csv	1.67M	Prior Home Credit applications
installments_payments.csv	13.6M	Instalment payment history
credit_card_balance.csv	3.84M	Revolving credit card monthly snapshots
POS_CASH_balance.csv	10.0M	POS loan / cash advance history

Why Home Credit Is the Better Primary Spine

- 10x more applicants than Taiwan Default (307k vs 30k)
- 7 relational side-tables enable deep feature engineering (instalment ratios, bureau aggregates, POS DPD)
- Real external credit bureau signals (EXT_SOURCE_1/2/3) as primary risk drivers
- 8.07% base rate matches real consumer-loan portfolios (vs 22% for Taiwan credit cards)
- SHA256 provenance verified — reproducible public download

Public-Data Limitations

- No temporal ordering — out-of-time split not possible
- Approved applicants only — reject inference unaddressed (MNAR selection bias)
- Single geography — no cross-market generalisation claim
- No real protected-class labels — fairness audit limited to proxies
- No income field — FOIR is a proxy (credit amount / goods price)

3. Architecture Overview

Stage	Description	Status	Artifact
Data audit	8-table row counts, DR, scale_pos_weight, temporal feasibility	[IMPLEMENTED]	g9a_home_credit_data_audit.json
Feature factory	140 features from 7 tables: ratios, aggregates, composites	[IMPLEMENTED]	g9a_feature_factory_report.json
Leakage audit	10/10 checks: TARGET exclusion, val-only fit, no test contamination	[IMPLEMENTED]	gold_pass1_leakage_audit.json
Train/val/test split	Stratified 60/20/20, seed=42, DR=0.0807 preserved	[IMPLEMENTED]	g9a_splits.pkl
Baseline tournament	12 models, 2 hard-failures, provisional champion CatBoost	[IMPLEMENTED]	g9a_model_tournament_report.json
Hyperparameter tuning	Optuna TPE, 5 models, validation-only, 4-8 trials	[IMPLEMENTED]	gold_pass2_tuning_trace.json
Calibration	Isotonic regression, val-only fit; numpy serving (no sklearn)	[IMPLEMENTED]	gold_pass2_calibration_report.json
Champion selection	9-component composite; LightGBM_mono wins	[IMPLEMENTED]	gold_pass2_champion_selection_report.json
Final test evaluation	Single evaluation on 61,503-row test set	[IMPLEMENTED]	gold_pass2_final_untouched_test_report.json
Score bands	GREEN/AMBER/RED; PD-semantic; cost-sensitive	[IMPLEMENTED]	gold_pass3_threshold_scoreband_report.json
SHAP reason codes	pred_contrib; 20 global + 4 local cases; ECOA drafts	[IMPLEMENTED]	gold_pass3_shap_reason_code_report.json
Reason-code stability	30 bootstraps; top-5 features 30/30 stable	[IMPLEMENTED]	gold_pass3_reason_code_stability.json
Fairness/proxy audit	Age, income, employment, region proxy groups	[PARTIALLY VERIFIED]	gold_pass3_fairness_proxy_audit.json
Drift/PSI baseline	val-vs-test PSI=0.0002; feature PSI STABLE	[PARTIALLY VERIFIED]	gold_pass3_drift_vintage_stress.json
RAG/LLM governance	6-case BM25+LLM demo, abstain, ASSISTIVE_ONLY	[SIMULATED]	gold_pass3_rag_llm_demo_report.json
GP5 LSTM experiment	1-layer LSTM encoder on installments; challenger AUC=0.7264	[IMPLEMENTED]	gp5_augment_and_retrain.py

4. Feature Engineering [IMPLEMENTED]

140 features engineered from 7 source tables after constant/near-zero-variance removal. All imputation and transformations fit on train only and applied to val/test.

Application Table — Core Ratio Features

Feature	Definition	Monotone
CREDIT_TO_INCOME	Total credit / reported income (FOIR proxy)	-
CREDIT_TO_GOODS	Credit amount / goods price (LTV proxy)	-
CREDIT_TO_ANNUITY	Credit amount / annual instalment (repayment capacity)	-
ANNUITY_TO_INCOME	Proposed annual repayment / income (DSR)	-
AGE_YEARS	Applicant age in years	-1 (older = lower risk)
EMPLOYED_YEARS	Employment tenure in years (sentinel fix applied)	-1
FLAG_EMPLOYED_ANOMALY	Binary: DAYS_EMPLOYED=365243 (not employed)	-

Bureau Aggregates (bureau + bureau_balance)

Feature	Definition	Monotone
BUREAU_ACTIVE_COUNT	Number of active bureau credit lines	-
BUREAU_OVERDUE_RATIO	Overdue accounts / total accounts	+1
BUREAU_AMT_OVERDUE	Total outstanding overdue balance	+1
BB_DPD_RATIO_MEAN	Mean DPD ratio across bureau balance records	+1

Previous Application + Instalment + Credit Card + POS Features

Feature	Source Table	Monotone
PREV_REFUSAL_RATE	previous_application	+1
PREV_AMT_CREDIT_MEAN	previous_application	-
INST_LATE_RATIO	installments_payments (rank-4 SHAP)	+1
CC_DPD_RATIO	credit_card_balance	+1
CC_ATM_RATIO	credit_card_balance	-
POS_IS_DPD_RATIO	POS_CASH_balance	+1

Composite Feature

BEHAVIORAL_RISK_SCORE = 0.4 x INST_LATE_RATIO + 0.3 x CC_DPD_RATIO + 0.2 x POS_IS_DPD_RATIO + 0.1 x BUREAU_OVERDUE_RATIO

Remaining ~120 features: one-hot encoded categoricals (NAME_CONTRACT_TYPE, CODE_GENDER, etc.), raw numeric fields from application_train (AMT_GOODS_PRICE, EXT_SOURCE_1/2/3, OWN_CAR_AGE, etc.), and additional aggregates from side-tables. All passed through LightGBM which handles missing values natively.

Leakage Controls

- TARGET excluded from all feature sets — confirmed by 10/10 pre-tuning leakage audit
- Post-outcome bureau fields reviewed; DAYS_DECISION in side-tables controlled
- Group leakage: 0 SK_ID_CURR overlap between train and test (confirmed by GP1 audit check #5)
- Calibrator fit on val only — never on train, never on test

5. Model Tournament

Baseline Tournament — G9A [IMPLEMENTED]

12 models run with near-default hyperparameters as a pre-tuning audit. Purpose: establish competitive ordering and confirm leakage-free pipeline, not find the best model.

Model	Val AUC	Status
CatBoost + Platt	0.7716	Provisional baseline champion
XGBoost + Platt	0.7703	Contender
LightGBM_base + Platt	0.7203	BUG: early stopping fires at iteration 1 with scale_pos_weight=11.39
LightGBM_monotonic + Platt	0.7203	Same bug as LightGBM_base
Random Forest	0.70	REJECTED — insufficient AUC on 140-feature space
Logistic Regression	0.65	REJECTED — linear boundary insufficient
TabNet	HARD_FAIL	CPU ~6 min/epoch; estimated 400-800h on 184k rows
sklearn GBM	HARD_FAIL	Wall-time exceeded on CPU sandbox

Tuned Tournament — Gold Pass 2 [IMPLEMENTED]

5 models tuned with Optuna TPE. LightGBM early-stopping bug fixed: n_estimators treated as a search parameter (200-1000 range), early stopping removed.

Model	Val AUC	Composite Score	Role
LightGBM_monotonic	0.7734	0.7312	CHAMPION + GOVERNANCE
XGBoost_monotonic	0.7699	0.7294	Contender
LightGBM_base	0.7724	0.6811	Contender (no governance premium)
CatBoost	0.7708	0.6802	Contender (no monotone constraints)
XGBoost	0.7704	0.6801	Contender

Early Stopping Bug — Root Cause

With eval_metric="auc" and early_stopping_rounds=50 on scale_pos_weight=11.39 imbalanced data, the first tree dominates AUC improvement. Subsequent marginal gains fall below tolerance. Training halts at 1-9 trees. Model looks normal but is severely undertrained. Fix: remove early stopping; treat n_estimators as Optuna hyperparameter.

6. Hyperparameter Tuning [IMPLEMENTED]

Field	Value
Algorithm	Optuna Tree-structured Parzen Estimator (TPE)
Seed	42
Budget	35-second wall-clock per model (CPU-only sandbox, 44s bash limit)
Trial counts	LightGBM_base: 6 | LightGBM_mono: 5 | CatBoost: 8 | XGBoost: 4 | XGBoost_mono: 4
Champion selection	Validation set only — test set never touched during HPO
Calibration	Fit on val only — after HPO, before test evaluation
Test evaluation	Exactly once, after champion locked

Search Space (LightGBM_monotonic)

Hyperparameter	Range	Best Value
n_estimators	200-1000	Best trial value (see tuning_trace.json)
num_leaves	31-255	TPE-selected
learning_rate	0.01-0.3 (log)	TPE-selected
max_depth	3-12	TPE-selected
subsample	0.5-1.0	TPE-selected
colsample_bytree	0.5-1.0	TPE-selected
reg_lambda (L2)	1e-3 to 10 (log)	TPE-selected
reg_alpha (L1)	1e-3 to 10 (log)	TPE-selected
min_child_samples	10-100	TPE-selected

Interview note: Trial counts (4-8) are lower than the 100-trial plan in the tuning spec. CPU sandbox with 44s bash limit is the constraint. The planning document specifies 100 trials; actual counts are documented in gold_pass2_tuning_trace.json. Results represent a reasonable local optimum. On GPU with 100 trials: estimated 0.001-0.003 further AUC improvement.

7. Champion Model [IMPLEMENTED]

LightGBM with Monotone Constraints + Isotonic Calibration

Field	Value
Model family	LightGBM Gradient Boosted Trees (leaf-wise growth, histogram splits)
Variant	15 monotone constraints — 12 at +1 (risk-increasing), 3 at -1 (risk-decreasing)
Calibration	Isotonic regression (val-only fit), served as numpy interp — zero sklearn dependency
Composite score	0.7312 (9-component selection)

Performance Metrics

Metric	Val	Test (Final, Untouched)	Interpretation
AUC	0.7734	0.7769	P(defaulter ranks above non-defaulter)
PR-AUC	0.2661	0.2628	3.3x random baseline at 8% DR
KS	0.4121	0.4141	Max CDF separation; >0.40 is strong
ECE	0.0051	0.0034	Near-production calibration quality
Brier	-	0.0668	Mean squared error of probability predictions

9-Component Composite Score Breakdown

Component	Weight	LightGBM_mono	CatBoost
Val AUC (normalised)	25%	0.7734	0.7708
Val PR-AUC (normalised)	15%	0.2661	0.2578
KS statistic	15%	0.4121	0.4094
ECE (inverted)	15%	0.0051	0.0098
Explainability (SHAP+mono)	10%	1.00	0.50
Adverse-reason readiness	5%	1.00	0.50
Inference latency	5%	fast	slower
Calibration method quality	5%	isotonic	platt only
Governance premium (monotone)	5%	YES	NO
COMPOSITE SCORE	100%	0.7312	0.6802

Why Monotone Constraints Matter

Monotone constraints guarantee directional interpretability auditable without per-applicant SHAP. A credit officer can verify: "for any two applicants identical in all features except BUREAU_OVERDUE_RATIO, the one with the higher ratio will always receive a higher predicted default probability." This is an SR 26-2-aligned interpretability property. Cost: some split flexibility. In practice, LightGBM_mono AUC (0.7734) exceeds LightGBM_base (0.7724) — constraints act as effective regularisation on the 140-feature space.

8. Calibration [IMPLEMENTED]

Step	Detail
Calibrator	IsotonicRegression(out_of_bounds="clip") — piecewise-monotone interpolation
Fit data	Validation set only
Platt	Evaluated but not selected — sigmoid slope/intercept produced higher test ECE
Test ECE	0.0034 — near-production-quality calibration
Serving	Extracted as iso_x.npy and iso_y.npy — np.interp(raw_prob, iso_x, iso_y) — zero sklearn dependency
Raw GBM ECE	0.296 (uncalibrated) — calibration reduces ECE by ~98%

Calibration Forensics Discovery

The GP2 training report originally stated "Platt selected." Forensic inspection of champion_calibrated.pkl confirmed: cal["selected"] = "isotonic". The isotonic calibrator was actually selected at training time. The training report was wrong. When initial serving code used the Platt formula, calibrated probabilities diverged from training by up to 0.53 — a clear red flag caught during Cloud Run deployment.

The Numpy Serving Solution

iso_x = iso.X_thresholds_ # shape (134,) — raw probability thresholds iso_y = iso.y_thresholds_ # shape (134,) — calibrated probability values # At serve time — exact replication, zero sklearn: calibrated_prob = float(np.interp(raw_prob, iso_x, iso_y))

np.interp implements the same piecewise-linear interpolation that IsotonicRegression.predict() uses. Max difference between the two: 0.0 across all test values. This eliminates sklearn version lock at serve time entirely.

Why Calibrated PD Matters

- A raw GBM score is not a probability — it is an uncalibrated log-odds output
- After isotonic calibration, PD=0.20 means the model genuinely estimates a 20% default probability
- Enables semantically defensible score-band thresholds
- Enables the cost-sensitive threshold formula: theta* = C_reject / (C_bad + C_reject)
- ECE=0.0034 is a measurable governance metric, traceable to artifact

9. Thresholds & Score Bands [IMPLEMENTED]

Band	Threshold	Test %	Test DR	Policy
GREEN	PD < 0.20	89.72%	5.77%	Auto-approve eligible (subject to hard-stop check)
AMBER	0.20 <= PD < 0.40	9.80%	26.96%	Manual credit officer review required
RED	PD >= 0.40	0.47%	53.77%	Enhanced underwriting or decline

Threshold Justification

PD=0.20 semantic justification: At this boundary the model estimates 20% default probability. The GREEN band observed DR (5.77%) is materially below 20%, confirming conservative assignment. Defensible to auditor: "applicants below this line have a predicted default probability under 20%."

Cost-sensitive analysis [SIMULATED]: Under scenario economics (C_bad=10, C_reject=1, C_review=0.3), the cost-optimal single threshold is 0.47 with expected loss 0.0807/applicant. These are scenario assumptions, not real bank economics. Operational threshold requires real LGD, NIM, and regulatory capital parameters.

Three Threshold Methods Evaluated

Method	Threshold	Approval Rate	DR	Assessment
KS-optimal	0.0961	50.7%	5.6%	Too aggressive for practical policy
Cost-optimal (C_bad=10, C_reject=1)	0.47	~99%	~8%	Scenario-assumed; not real bank econ
PD-semantic (CHOSEN)	0.20 / 0.40	89.7% / 9.8% / 0.5%	Banded	Interpretable; defensible to auditor

10. SHAP Reason Codes & Stability [IMPLEMENTED]

Method

LightGBM built-in pred_contrib=True (native SHAP TreeExplainer implementation). Avoids external SHAP library — which had a pandas index compatibility issue with non-default-indexed DataFrames. Native implementation is equivalent for tree models and zero additional dependencies.

Global Feature Importance (mean |SHAP|, 1000-sample draw)

Rank	Feature	Mean |SHAP|	Monotone	Human Label
1	EXT_SOURCE_MEAN	0.510	-1	External credit bureau composite score
2	CREDIT_TO_ANNUITY	0.141	N/A	Credit amount vs annual repayment capacity
3	CREDIT_TO_GOODS	0.139	N/A	Loan-to-goods-value ratio
4	INST_LATE_RATIO	0.129	+1	Proportion of late instalment payments
5	EXT_SOURCE_1	0.120	N/A	External bureau score 1
6	OWN_CAR_AGE	0.092	N/A	Vehicle age (ownership proxy)
7	EMPLOYED_YEARS	0.091	-1	Employment tenure
8	BUREAU_ACTIVE_COUNT	0.087	N/A	Active bureau credit lines
9	AMT_GOODS_PRICE	0.082	N/A	Goods price of financed item
10	EXT_SOURCE_3	0.078	N/A	External bureau score 3

Local Cases Computed

Case	PD	Band	Primary Driver	SHAP Value
GREEN_APPROVE	0.0335	GREEN	EXT_SOURCE_MEAN	-0.829
AMBER_REVIEW	0.2056	AMBER	EXT_SOURCE_MEAN	+0.939
RED_HIGH_RISK	0.4164	RED	EXT_SOURCE_MEAN	+1.300
AMBER_CONFLICT	0.3475	AMBER	EXT_SOURCE_MEAN	+1.232

Bootstrap Stability (30 Iterations x 500-row Samples)

Feature	Top-5 Presence	Rank-1 Count	Stability Tier
EXT_SOURCE_MEAN	30/30	30/30	HIGH
EXT_SOURCE_1	30/30	-	HIGH
CREDIT_TO_GOODS	30/30	-	HIGH
CREDIT_TO_ANNUITY	30/30	-	HIGH
INST_LATE_RATIO	30/30	-	HIGH

EXT_SOURCE_MEAN is rank-1 in all 30 bootstraps. Top-5 reason-code set is robust to sampling variation — suitable for production adverse-action drafting (subject to officer review).

Adverse-Action Language (Draft, ASSISTIVE_ONLY)

- EXT_SOURCE_MEAN: "External credit bureau composite score below threshold"
- INST_LATE_RATIO: "High proportion of late instalment payments"
- CREDIT_TO_ANNUITY: "Credit amount high relative to annual repayment capacity"
- CREDIT_TO_GOODS: "Loan-to-goods-value ratio elevated"

These are draft aids, not legally compliant adverse action notices. ECOA/Regulation B requires reason codes from an approved list, reviewed by a licensed credit officer.

11. Fairness / Proxy Audit [PARTIALLY VERIFIED]

Proxy-variable analysis on 61,503-row test set. No protected-class labels available in Home Credit. Scope explicitly: skeleton, not certification.

Proxy Group	Approval Rate	Observed DR	Assessment
Age < 30	80.5%	11.5%	DR differential explains approval gap
Age 30-50	88.6%	8.8%	Baseline cohort
Age 50+	95.5%	5.6%	Lower DR explains higher approval
Income low tercile	89.0%	8.1%	Narrow spread across terciles
Income mid tercile	88.5%	8.8%	Baseline
Income high tercile	91.3%	7.5%	Slight improvement with higher income
Employed	88.3%	8.7%	Normal
Not-employed proxy	96.2%	5.2%	Sentinel encodes retired/students — lower DR
Region 1 (low-risk)	96.5%	4.9%	DR differential explains gap
Region 3 (high-risk)	81.7%	11.2%	DR differential explains gap

What This Proves vs Does Not Prove

What this proves: No evidence of model amplifying approval-rate disparities beyond base-rate differences. Model appears to differentiate risk, not proxy demographics.

What it does NOT prove: Not a fairness certification. Disparate Impact ratio not computable (no protected-class labels). A full ECOA/fair-lending review would require demographic enrichment and independent analysis.

12. Drift / PSI Baseline [PARTIALLY VERIFIED]

Comparison	PSI	Interpretation
val vs test	0.0002	STABLE — key operational metric
train vs val	~8.10	Platt/isotonic calibration artefact (not a drift signal)

The train PSI is explained by the isotonic calibrator being fit on val-set scores. The calibrated val/test distribution is consistent (PSI=0.0002). All top-10 feature PSIs between train and test are below 0.001.

AUC by Split

Split	AUC	Interpretation
Train	0.8526	In-sample (expected overfitting)
Val	0.7734	Out-of-sample estimate
Test	0.7769	Final untouched evaluation (better than val — within noise)

Production Monitoring Policy (DOCUMENTED)

PSI Range	Status	Action	SLA
PSI < 0.10	STABLE	No action required	-
0.10 - 0.25	SLIGHT_SHIFT	Model owner review	30 days
PSI > 0.25	SIGNIFICANT_SHIFT	MRC escalation	5 business days

- If monthly KS drops > 0.05 from deployment baseline (0.4141): emergency model review regardless of PSI
- Approve-zone observed DR WARN at 2x baseline (10%), ALERT at 25%, CRITICAL at 35%

True vintage validation [FUTURE]: Requires timestamped application data. Home Credit has none. Monthly PSI monitoring on score distribution + top-10 features is the production monitoring recommendation.

13. RAG / LLM Governance Assistant [SIMULATED]

Architecture

Component	Implementation
Retriever	BM25 over 5-document policy corpus
Abstain threshold	BM25 top-score < 0.25 — no output (prevents hallucinated policy citations)
LLM role	Draft memo and adverse-action language from retrieved policy section
Hard rule	LLM NEVER autonomously approves or rejects applicant in any case
All outputs	ASSISTIVE_ONLY + HUMAN_REVIEW_REQUIRED + NOT_FINAL_DECISION

6-Case Demonstration

Case	Band	Retrieved Policy	LLM Output	Abstain
GREEN_APPROVE	GREEN	POL-001 (Score Band Policy)	Approval eligibility memo	No
AMBER_REVIEW	AMBER	POL-001	Review memo + top-3 SHAP drivers	No
RED_DECLINE	RED	POL-002 (Adverse Action Policy)	Adverse-action draft + 3 reason codes	No
CONFLICT_OVERRIDE	AMBER	POL-004 (Override Protocol)	Override checklist; officer authority noted	No
OUT_OF_DOMAIN	N/A	None (BM25=0.0)	ABSTAIN — no output generated	YES
DRIFT_ALERT	MONITOR	POL-003 (Monitoring Policy)	MRC escalation memo	No

What the LLM supports: Draft memos, reason-code language, policy lookups, checklist generation, escalation flagging.

What the LLM NEVER does: Make or suggest credit approval/rejection, claim legal compliance, access live applicant data, produce output when BM25 < 0.25.

14. Evidence Artifacts Ledger

Artifact	Key Numbers	Claim Enabled
gold_pass1_leakage_audit.json	10/10 PASS, safe_to_tune=true	No target or post-outcome leakage
gold_pass1_data_spine_validation.json	21/21 PASS, 0 ID overlap	No data leakage between splits
gold_pass2_tuning_trace.json	5 models, 4-8 trials	Validation-only Optuna HPO
gold_pass2_calibration_report.json	Isotonic ECE_val=0.0051	Isotonic calibration selected
gold_pass2_champion_selection_report.json	Composite=0.7312	9-component composite champion
gold_pass2_final_untouched_test_report.json	AUC=0.7769, ECE=0.0034	Final test: AUC=0.7769
gold_pass3_threshold_scoreband_report.json	GREEN 89.7%, AMBER 9.8%, RED 0.5%	Score-band policy
gold_pass3_shap_reason_code_report.json	EXT_SOURCE_MEAN=0.510, 4 local cases	SHAP reason codes
gold_pass3_reason_code_stability.json	30/30 top-5, 30/30 rank-1	Reason codes stable across bootstraps
gold_pass3_fairness_proxy_audit.json	4 proxy groups, aligned with DR	Fairness proxy audit skeleton
gold_pass3_drift_vintage_stress.json	PSI=0.0002, all features STABLE	val-vs-test PSI=0.0002 STABLE
gold_pass3_rag_llm_demo_report.json	6 cases, abstain fires	RAG/LLM 6-case governance demo
pulseguard_final_gold_audit.json	89.3%, GOLD	PulseGuard is GOLD checkpointed
docs/PULSEGUARD_MODEL_CARD.md	All lifecycle fields	Complete model card
04_EVIDENCE_LEDGER.md	All claims traced	Every claim has an artifact
06_CLAIM_BOUNDARY.md	Safe/forbidden per gate	Claim boundary maintained

14b. Feature Catalogue — All 140 Features by Source Table

application_train.csv — Base Features

Feature	Type	Monotone	Description
CREDIT_TO_INCOME	ratio	-	AMT_CREDIT / AMT_INCOME_TOTAL — debt-to-income proxy
CREDIT_TO_GOODS	ratio	-	AMT_CREDIT / AMT_GOODS_PRICE — LTV proxy
CREDIT_TO_ANNUITY	ratio	-	AMT_CREDIT / AMT_ANNUITY — repayment capacity
ANNUITY_TO_INCOME	ratio	-	AMT_ANNUITY / AMT_INCOME_TOTAL — debt service ratio
AGE_YEARS	transformed	-1	DAYS_BIRTH / -365 — older applicants lower risk
EMPLOYED_YEARS	transformed	-1	DAYS_EMPLOYED / -365; sentinel-corrected
FLAG_EMPLOYED_ANOMALY	binary	-	1 if DAYS_EMPLOYED=365243 (not employed encoding)
EXT_SOURCE_MEAN	aggregate	-1	Mean of EXT_SOURCE_1/2/3 — rank-1 SHAP driver
EXT_SOURCE_1	raw	-	External bureau score 1
EXT_SOURCE_2	raw	-	External bureau score 2
EXT_SOURCE_3	raw	-	External bureau score 3
AMT_CREDIT	raw	-	Loan amount requested
AMT_INCOME_TOTAL	raw	-	Applicant annual income
AMT_ANNUITY	raw	-	Proposed annual loan payment
AMT_GOODS_PRICE	raw	-	Price of financed goods
OWN_CAR_AGE	raw	-	Age of car owned (null if no car)
DAYS_REGISTRATION	raw	-	Days since current registration
DAYS_ID_PUBLISH	raw	-	Days since ID was published/changed
CNT_FAM_MEMBERS	raw	-	Number of family members
CNT_CHILDREN	raw	-	Number of children
REGION_RATING_CLIENT	raw	+1	Client region rating — higher is riskier
REGION_RATING_CLIENT_W_CITY	raw	+1	City-adjusted region rating
NAME_CONTRACT_TYPE	onehot	-	Cash loan vs revolving
CODE_GENDER	onehot	-	M/F encoding
FLAG_OWN_CAR	binary	-	Owns a car
FLAG_OWN_REALTY	binary	-	Owns real estate
NAME_INCOME_TYPE	onehot	-	Working/Pensioner/etc.
NAME_EDUCATION_TYPE	onehot	-	Education level
NAME_FAMILY_STATUS	onehot	-	Marital status
NAME_HOUSING_TYPE	onehot	-	Housing situation
OCCUPATION_TYPE	onehot	-	Occupation category
FLAG_WORK_PHONE / FLAG_PHONE / FLAG_EMAIL	binary	-	Contact flags
REG_REGION_NOT_LIVE_REGION	binary	+1	Address vs residence mismatch
LIVE_CITY_NOT_WORK_CITY	binary	+1	Lives and works in different cities
DAYS_LAST_PHONE_CHANGE	raw	-	Recency of phone number change

bureau.csv + bureau_balance.csv — Credit Bureau Features

Feature	Monotone	Description
BUREAU_ACTIVE_COUNT	-	Number of currently active credit bureau accounts
BUREAU_CLOSED_COUNT	-	Number of closed bureau accounts
BUREAU_TOTAL_COUNT	-	Total number of bureau accounts (active + closed)
BUREAU_OVERDUE_RATIO	+1	Overdue accounts / total bureau accounts
BUREAU_AMT_CREDIT_SUM	-	Sum of all bureau credit amounts
BUREAU_AMT_CREDIT_DEBT	+1	Total outstanding bureau debt
BUREAU_AMT_OVERDUE	+1	Total overdue balance across bureau accounts
BUREAU_CREDIT_TYPE_COUNT	-	Count of distinct credit types in bureau
BUREAU_MAX_OVERDUE_AMT	+1	Maximum single overdue amount in bureau
BUREAU_AVG_DAYS_CREDIT	-	Average age of bureau credit lines (days)
BB_DPD_RATIO_MEAN	+1	Mean DPD ratio across bureau balance monthly records
BB_DPD_RATIO_MAX	+1	Maximum DPD ratio in bureau balance history
BB_STATUS_ACTIVE_RATIO	-	Fraction of bureau balance records with active status
BB_STATUS_DPD_RATIO	+1	Fraction of bureau balance records showing DPD

previous_application.csv — Prior Application Features

Feature	Monotone	Description
PREV_APP_COUNT	-	Total number of prior Home Credit applications
PREV_APPROVED_COUNT	-	Number of prior approved applications
PREV_REFUSED_COUNT	+1	Number of prior refused applications
PREV_REFUSAL_RATE	+1	PREV_REFUSED_COUNT / PREV_APP_COUNT — rank-signal feature
PREV_AMT_CREDIT_MEAN	-	Mean credit amount in prior applications
PREV_AMT_ANNUITY_MEAN	-	Mean annuity in prior applications
PREV_DAYS_DECISION_MEAN	-	Mean days since prior application decision
PREV_PRODUCT_TYPE_CASH_RATIO	-	Fraction of prior apps for cash loans
PREV_CNT_PAYMENT_MEAN	-	Mean contracted payment count in prior loans

installments_payments.csv — Payment Behaviour Features

Feature	Monotone	Description
INST_LATE_RATIO	+1	Late payments / total payments — rank-4 global SHAP feature
INST_PAY_RATIO	-	Mean (actual_payment / scheduled_payment) — underpayment proxy
INST_DPD_MEAN	+1	Mean days past due across all instalment records
INST_DPD_MAX	+1	Maximum DPD in instalment history
INST_PAYMENT_DIFF_MEAN	+1	Mean shortfall between scheduled and actual payment
INST_AMT_INSTALMENT_SUM	-	Total instalment amount across all payments
INST_NUM_INSTALMENT_MEAN	-	Mean instalments per prior loan
INST_OVERPAY_RATIO	-	Ratio of overpayments in instalment history

credit_card_balance.csv + POS_CASH_balance.csv — Revolving/POS Features

Feature	Source	Monotone	Description
CC_DPD_RATIO	credit_card_balance	+1	Credit card days-past-due ratio across monthly records
CC_ATM_RATIO	credit_card_balance	-	ATM withdrawal / total credit card drawings
CC_BALANCE_TO_LIMIT	credit_card_balance	+1	Mean credit utilisation ratio on credit cards
CC_PAYMENT_RATIO	credit_card_balance	-	Mean actual / min payment ratio on cards
CC_OVERDUE_COUNT	credit_card_balance	+1	Count of months with credit card DPD > 0
POS_IS_DPD_RATIO	POS_CASH_balance	+1	POS instalment days-past-due ratio
POS_CNT_INSTALMENT_MEAN	POS_CASH_balance	-	Mean contracted instalment count (POS)
POS_MONTHS_BALANCE_MEAN	POS_CASH_balance	-	Mean months in POS balance history
POS_COMPLETED_RATIO	POS_CASH_balance	-	Fraction of POS contracts marked completed

Composite / Engineered Cross-Table Features

Feature	Formula	Rationale
BEHAVIORAL_RISK_SCORE	0.4*INST_LATE_RATIO + 0.3*CC_DPD_RATIO + 0.2*POS_IS_DPD_RATIO + 0.1*BUREAU_OVERDUE_RATIO	Weighted delinquency composite across all payment domain
EXT_SOURCE_MEAN	(EXT_SOURCE_1 + EXT_SOURCE_2 + EXT_SOURCE_3) / 3	Average external bureau signal — rank-1 SHAP globally
INCOME_PER_PERSON	AMT_INCOME_TOTAL / max(CNT_FAM_MEMBERS, 1)	Per-capita income proxy accounting for household size
PAYMENT_RATE	AMT_ANNUITY / AMT_CREDIT	Fraction of credit paid annually — repayment velocity

Remaining features to reach 140 total: additional one-hot encoded categoricals from application_train (NAME_CONTRACT_TYPE, WEEKDAY_APPR_PROCESS_START, ORGANIZATION_TYPE with 55 categories, etc.), binary flags (FLAG_DOCUMENT_*), and raw numeric fields not listed above. All 140 passed to LightGBM which handles missing values natively in histogram-based split-finding.

14c. Model Card (Inline Summary)

Field	Value
Model name	PulseGuard Champion — LightGBM_monotonic v2 (GP2)
Version	Gold Pass 2 — frozen 2026-06-18
Model type	Gradient Boosted Trees (LightGBM) with 15 monotone constraints
Calibration	Isotonic regression, val-only fit, served as numpy interp arrays
Training data	Home Credit Default Risk — 184,506 applicants (60% train split, seed=42)
Validation data	Home Credit Default Risk — 61,502 applicants (20% val split)
Test data	Home Credit Default Risk — 61,503 applicants (20% test split, evaluated once)
Target variable	TARGET (1=default within loan term, 0=repaid) — 8.07% positive rate
Features	140 features from 7 relational tables; all imputation fit on train only
Intended use	Portfolio demonstration of credit-risk model governance methodology
Out-of-scope use	Production lending decisions, regulatory submissions, live underwriting
Test AUC	0.7769
Test PR-AUC	0.2628
Test KS	0.4141
Test Brier	0.0668
Test ECE	0.0034
Composite score	0.7312 (9-component)
Governance audit	GOLD — 89.3% (134/150) on 15-dimension audit
SR 26-2 alignment	Designed to demonstrate SR 26-2 principles; NOT certified by regulatory body
Fairness status	Proxy audit only — no protected-class labels; NOT fairness-certified
Reject inference	NOT addressed — model trained on approved applicants only (MNAR bias)
Temporal validation	NOT possible — Home Credit dataset has no application timestamps
Deployment status	NOT deployed — champion pkl incompatible with Python 3.9 deployment environment
Champion selected	2026-06-18, by 9-component composite score
Model frozen	2026-06-18 — no retuning, no champion changes permitted after Gold checkpoint

15. Failure Modes & Mitigations

#	Failure	Detection	Gap
1	Target leakage	10/10 GP1 leakage audit — safe_to_tune=true	No automated re-audit if features change
2	Test-set leakage	Test evaluated exactly once, after all decisions locked	Manual discipline required
3	Post-outcome variables	GP1 temporal checks on side-tables	Future additions require re-audit
4	Temporal overclaim	FEASIBILITY_LIMITED documented; no OOT claim made	True OOT validation requires timestamped data
5	AUC-only selection	9-component composite; governance premium for monotone	None
6	Calibration overfit	Isotonic val ECE evaluated; test ECE=0.0034 is real measure	None
7	SHAP instability	30-bootstrap stability check; top-5 features 30/30	Full corpus SHAP would be stronger
8	Near-threshold instability	AMBER_CONFLICT local case computed at PD=0.347	Fine-grained boundary stability unmeasured
9	Fairness overclaim	SKELETON label; proxy analysis only; forbidden claims documented	No protected-class labels
10	Cost matrix overclaim	SCENARIO_ASSUMPTIONS label; interview answer prepared	Requires real LGD/NIM data
11	Drift hidden by random split	val-vs-test PSI=0.0002 is correct; train PSI explained	No true temporal drift test possible
12	RAG wrong policy retrieval	Abstain threshold BM25<0.25; OOD cases abstain correctly	Small corpus may have weak coverage
13	LLM makes credit decision	Hard rule: LLM never approves/rejects in any demo case	Ongoing prompt discipline in production
14	Adverse-action legal overclaim	ASSISTIVE_ONLY + HUMAN_REVIEW_REQUIRED on all outputs	Licensed officer must review
15	Public-data-to-production overclaim	PORTFOLIO PROJECT stated throughout; Model card explicit	MNAR bias unaddressed

16. Claim Boundary

Locked Safe Claims

- Tuned LightGBM_monotonic + Isotonic: AUC=0.7769, PR-AUC=0.2628, KS=0.4141, ECE=0.0034 on 61k held-out test
- Validation-only Optuna HPO; champion selected by 9-component composite score
- Monotone constraints on 15 features support SR 26-2 directional interpretability
- EXT_SOURCE_MEAN rank-1 SHAP driver in 30/30 bootstraps; top-5 features stable 30/30
- Score-band policy: GREEN<0.20 (89.7%, DR=5.8%), AMBER 0.20-0.40 (9.8%, DR=27%), RED>=0.40 (0.5%, DR=54%)
- RAG/LLM: abstains on OOD (BM25<0.25); LLM never makes credit decisions in any of 6 demo cases
- Fairness proxy: approval-rate differentials aligned with DR differentials; no amplification
- val-vs-test PSI=0.0002; all top-10 feature PSIs STABLE
- PulseGuard scores GOLD at 89.3% (134/150) on 15-dimension governance audit
- FastAPI endpoint at Cloud Run serves G4 XGBoost demo (AUC=0.6261, synthetic); GP2 champion NOT deployed due to sklearn version incompatibility — serving gap fully documented

Locked Forbidden Claims

- Production lending system or live deployment
- SR 26-2 certified or regulatory compliance certified
- Legally compliant adverse action notice
- Fairness certified or disparate impact compliant
- Real bank data or real lender economics
- Full vintage/out-of-time validation
- LLM makes or influences credit decisions
- 100 Optuna trials completed — actual: 4-8 per model
- CatBoost is the final champion — champion is LightGBM_monotonic
- AUC=0.7716 is the tuned result — that is the BASELINE untuned CatBoost score
- Live endpoint AUC=0.7769 — endpoint serves G4 demo; AUC=0.6261 on synthetic data

Resume-Safe Wording

"Built end-to-end credit-risk ML pipeline on Home Credit Default Risk (307k applicants, 57.4M rows); champion LightGBM + Isotonic achieves AUC=0.7769, KS=0.41, ECE=0.0034 — selected via 9-component composite score including monotone constraints, calibration, and adverse-reason readiness."

17. Interview Q&A; Master Deck

Standard Questions

Q1 — 60-second opener: "Tell me about PulseGuard in one minute."

PulseGuard is my credit-risk governance portfolio project built on the Home Credit Default Risk dataset — 307,000 applicants, 57 million rows across 7 relational tables. I engineered 140 features, ran a 12-model baseline tournament, then tuned 5 models with Optuna. The champion is LightGBM with monotone constraints, calibrated with isotonic regression — AUC=0.77, ECE=0.003 on a 61,000-row holdout. I then added the governance layer: score-band policy, SHAP reason codes with bootstrap stability, a fairness proxy audit, a drift baseline, and a local RAG policy assistant that drafts adverse-action memos for credit officer review. Every claim is traceable to an evidence artifact. Not a production system — a demonstration of the full governed ML lifecycle.

Q2 — 2-minute walkthrough: "Walk me through the project."

I will cover it in four stages: data, model, governance, and boundaries. DATA: Home Credit Default Risk — 307,000 applicants at 8% default rate, across 7 tables including bureau history, instalment payments, credit cards, and POS cash records. I engineered 140 features: ratio features like credit-to-annuity and loan-to-goods-value, behavioural aggregates like late-payment ratio and bureau overdue ratio, and a composite behavioural risk score. 10-check leakage audit before any model training — test set never touched until final evaluation. MODEL: 12-model baseline tournament. LightGBM was underperforming due to an early-stopping bug on imbalanced data. Fixed and ran Optuna HPO on 5 models. Champion: LightGBM with monotone constraints, selected by a 9-component composite score. Isotonic calibration produces ECE=0.003 on test set. Final test AUC: 0.7769. GOVERNANCE: Three-zone score-band policy (GREEN/AMBER/RED), SHAP reason codes stable across 30 bootstrap resamples, fairness proxy audit on age/income/employment/region, PSI drift baseline, and a local BM25 policy RAG with an LLM that drafts adverse-action memos — ASSISTIVE_ONLY. BOUNDARIES: Portfolio project, not production. Limitations documented: no temporal validation, MNAR selection bias, proxy fairness only, 4-8 Optuna trials. Every claim in my resume has an artifact.

Q3 — "Why LightGBM_monotonic and not CatBoost?"

CatBoost val AUC=0.7708 vs LightGBM_mono=0.7734 — only a 0.0026 AUC gap. But the composite score is 0.6802 for CatBoost vs 0.7312 for LightGBM_mono. The composite includes a 10% explainability weight: LightGBM_mono gets a full score for SHAP + monotone constraints; CatBoost only gets partial credit. And a 5% weight for adverse-reason readiness — monotone constraints make every prediction auditable without SHAP. More importantly: LightGBM_mono is both the performance champion AND the governance champion. There is no trade-off to defend. That is a better story than "we traded 0.002 AUC for governance compliance."

Q4 — "Why monotone constraints?"

Monotone constraints enforce directional interpretability at every tree split. If BUREAU_OVERDUE_RATIO has a +1 constraint, then for any two applicants identical in all other features, the one with the higher overdue ratio will always receive a higher predicted default probability — guaranteed by the model's construction. This matters for SR 26-2 model risk management. An auditor can verify the directional claim by inspection without needing a SHAP explanation. A credit officer can explain the model's logic to a regulator without showing a SHAP waterfall chart. The cost is some flexibility in tree splits. In practice, LightGBM_mono AUC=0.7734 vs LightGBM_base AUC=0.7724 — the monotone version is actually slightly better on AUC, probably because the constraints act as useful regularisation on the 140-feature space.

Q5 — "How did you avoid leakage?"

Three layers. First, TARGET excluded from features — confirmed by 10/10 pre-tuning leakage audit. Second, the calibrator was fit on validation set only — not on train, not on test. Third, the test set was used exactly once, after champion was locked, calibration complete, and score-band thresholds defined. The audit also verified: no applicant ID overlap between splits, no post-outcome proxy variables in the side-tables. The test set was never consulted during Optuna HPO — all optimisation was on the validation set.

Q6 — "What is the early stopping bug?"

With eval_metric="auc" and early_stopping_rounds=50 on scale_pos_weight=11.39 imbalanced data, the first tree dominates the AUC improvement. Subsequent trees add only marginal gain. If those gains fall below the default tolerance, training halts at 1-9 trees. The model looks like it ran normally but is severely undertrained. Result: LightGBM G9A AUC=0.7203 — looked plausible, was wrong. I accepted it as BASELINE_NOT_TUNED at Gold Pass 1 without diagnosing the root cause. Gold Pass 2 fix: remove early stopping; treat n_estimators as Optuna search parameter (range 200-1000). After fix, LightGBM_mono reaches AUC=0.7734 and becomes the champion. This bug pattern appears in any imbalanced problem with high scale_pos_weight — production fraud models at 0.1% positive rate have this risk at scale_pos_weight=999.

Q7 — "How did tuning work?"

Optuna Tree-structured Parzen Estimator — a Bayesian search that models which hyperparameter configurations produce good validation AUC and proposes new configurations accordingly. I searched over n_estimators, learning rate, num_leaves, max_depth, subsample, colsample_bytree, and regularisation parameters. All tuning optimised on validation AUC — test set never consulted. Trial counts: 4-8 per model, not 100. CPU-only sandbox with 44-second bash execution limit. I am honest about that: the tuning plan specified 100 trials; actual counts are documented in gold_pass2_tuning_trace.json. On GPU with 100 trials I would likely get 0.001-0.003 further AUC improvement.

Q8 — "What is ECE?"

Expected Calibration Error. It measures how well predicted probabilities match observed default rates. A calibrated model with PD=0.20 should produce ~20% defaults in that bucket. ECE is computed by binning predictions into 10 buckets, averaging the absolute difference between mean predicted PD and observed DR, weighted by bucket size. Champion ECE=0.0034 on test set. Raw uncalibrated LightGBM ECE=0.296. Isotonic calibration reduces ECE by ~98%. ECE=0.0034 means the model's predicted probabilities are extremely close to observed outcomes — near-production-quality calibration.

Q9 — "What is KS and why does it matter for credit?"

KS (Kolmogorov-Smirnov) is the maximum separation between the cumulative distribution functions of predicted scores for defaulters and non-defaulters. KS=0.4141 means there is a 41.4 percentage-point gap between the two CDFs at the point of maximum separation. It is a useful single-number discriminative measure that does not depend on a specific threshold. For credit scoring, KS>0.40 is conventionally considered strong. KS is widely used in credit risk because lenders care about performance at a specific operating point; AUC is better for comparing models across all thresholds.

Q10 — "Why PR-AUC matters at 8% default rate?"

ROC-AUC rewards true negatives heavily at 8% DR — 11.4 non-defaulters per defaulter. A model mediocre at finding defaulters can still score well on ROC. PR-AUC focuses on precision and recall for the minority class. A random classifier at 8% DR has PR-AUC ~= 0.08 (the base rate). Champion PR-AUC=0.2628 is 3.3x the random baseline. PR-AUC is the honest measure of how well the model actually identifies defaulters.

Q11 — "How did you create reason codes?"

Using LightGBM's built-in pred_contrib=True parameter — the LightGBM implementation of SHAP TreeExplainer values. For each prediction, it returns the marginal contribution of each feature to the log-odds of default. I rank by absolute value and map the top drivers to human-readable language. EXT_SOURCE_MEAN contribution of +0.94 for an AMBER applicant maps to "External credit bureau composite score below threshold." INST_LATE_RATIO contribution of +0.45 maps to "High proportion of late instalment payments." I used LightGBM's built-in implementation rather than the external SHAP library because the external library had a pandas index compatibility issue with our non-default-indexed DataFrames. The native implementation is equivalent for tree models and avoids the dependency.

Q12 — "Are they legally compliant adverse action notices?"

No. ECOA and Regulation B require adverse action notices with specific reason codes drawn from an approved list, reviewed by a licensed credit officer. Our SHAP-derived language is a draft aid — it helps a credit officer identify the right reason codes faster. The output is always tagged ASSISTIVE_ONLY and HUMAN_REVIEW_REQUIRED. A credit officer must review, confirm, and sign off on any adverse action notice sent to an applicant. This boundary is documented in the model card, the governance report, the claim boundary document, and the RAG/LLM system's hard rules.

Q13 — "What are the biggest limitations?"

Five material ones. First, reject inference: model trained on approved applicants only — MNAR selection bias; declined population performance is unknown. Second, no temporal holdout: no timestamps in Home Credit; val and test are from the same distribution. Third, single geography: Eastern European portfolio — generalisation to other markets is untested. Fourth, CPU trial count: 4-8 Optuna trials instead of 100; reasonable local optimum, not globally optimal. Fifth, fairness: proxy audit only — protected-class labels unavailable. None of these are hidden — all documented in the model card.

Q14 — "How did you audit fairness?"

I ran a proxy audit — the maximum achievable without protected-class labels. Home Credit does not contain race, sex, or national origin fields. I used age bands, income terciles, employment status proxy, and regional rating as approximate proxies. For each group I computed approval rate at the GREEN band threshold (PD<0.20) and compared to the observed default rate for that group. Finding: all approval-rate differences are directionally aligned with default-rate differences. The model appears to differentiate risk, not discriminate by demographics. Explicitly labelled a governance skeleton, not a fairness certification.

Q15 — "I tested your live endpoint. AUC is 0.62. What is going on?"

Correct, and documented. The live endpoint at Cloud Run demonstrates the serving architecture: preprocessing pipeline, isotonic-calibrated probability, SHAP top-3 reason codes, score banding, and the ASSISTIVE_ONLY response structure. The model inside is the G4 demo model — XGBoost trained on 50,000 rows of synthetic data, 28 features, AUC=0.6261. The GP2 LightGBM champion hit a deployment blocker: pkl artifacts serialized with sklearn 1.7.2 require Python 3.10+; my deployment environment runs Python 3.9, which maxes out at sklearn 1.6.1. Cross-version pkl deserialization is not supported. The serving pattern is what the endpoint demonstrates. For champion quality, the evidence artifacts in outputs/evidence/ have every metric traceable to a specific gate.

Q16 — "How would you monitor drift in production?"

Monthly PSI on the score distribution and top-10 input features. PSI<0.10 is STABLE; 0.10-0.25 is SLIGHT_SHIFT (model owner review, 30-day SLA); >0.25 is SIGNIFICANT_SHIFT (MRC escalation, 5 business days). Additionally: if monthly KS drops more than 0.05 from the deployment baseline (0.4141), that triggers emergency model review regardless of PSI. Approve-zone observed DR: WARN at 2x baseline (10%), ALERT at 25%, CRITICAL at 35%. And the score distribution itself — if mean score shifts by more than 0.05, flag for investigation.

Q17 — "What would production require?"

Eight things at minimum: independent model validation team review (SR 26-2); out-of-time validation on recent vintages; real bank economics for the cost matrix (real LGD, NIM, regulatory capital); fairness audit with real demographic labels and legal review; reject inference correction for the approved-applicant selection bias; FastAPI serving with training-serving parity check (version-pinned sklearn/Dockerfile); monthly PSI monitoring with automated WARN/ALERT/CRITICAL escalation; credit officer workflow integration for adverse-action notice review and sign-off.

Q18 — "What would you improve next?"

Top three in order of impact. First, GPU with 100 Optuna trials — expect 0.001-0.003 AUC improvement and more robust champion. Second, temporal validation if I can find a timestamped public credit dataset — HMDA has application dates and would allow true out-of-time testing; that is the single biggest methodological gap. Third, a stronger fairness audit with demographic enrichment — finding or acquiring protected-class labels for Home Credit or a comparable dataset is the highest-governance-value next step.

Q19 — "What does the LLM do exactly?"

The LLM governance assistant has three functions: policy retrieval, memo drafting, and adverse-action language suggestion. It takes an anonymised case summary, retrieves the relevant policy section using BM25, and drafts a memo for the credit officer to review. The BM25 abstain threshold (0.25) prevents the LLM from generating output when no relevant policy is found — the out-of-domain query in Case 5 produces zero output, not a hallucinated policy citation. The LLM never generates a credit decision. In all six demo cases, every output is either a memo, a checklist, or a reason-code draft — all labelled ASSISTIVE_ONLY.

Q20 — "How would you explain PulseGuard to a non-technical hiring manager?"

The model estimates how likely each loan applicant is to default. Applicants are sorted into three buckets: Green for low risk, Amber for medium risk, and Red for high risk. Green applicants can be approved quickly — very low estimated default rate. Amber applicants go to a credit officer for manual review. Red applicants need extra scrutiny or are declined. When we decline an applicant or send them to review, the system drafts the top reasons — for example, "high proportion of late payments in prior credit history" — for the credit officer to review and confirm. The officer makes the final decision; the model helps them work faster and more consistently.

18. Adversarial Follow-Up Drill

Second-Punch Questions

Q — "You mentioned 57.4M rows. That is the raw input, right?"

Correct. Each side table is aggregated to applicant level via GROUP BY SK_ID_CURR — bureau_balance from 27M monthly records down to a handful of ratios per applicant. The final modeling matrix is 307,511 rows x 140 features. 57.4M is the raw data volume processed during feature engineering, not the size of what goes into the model.

Q — "What did you do with nulls?"

Median imputation for numerics, mode for categoricals, all fit on train only and applied to val and test. Explicit missingness indicator flags added for features where missingness is a signal — OWN_CAR_AGE is null when the applicant owns no car, which is informative. LightGBM handles missing values natively in splitting decisions, so imputation is belt-and-suspenders.

Q — "Why not k-fold cross-validation?"

Three reasons. Calibration discipline: Platt/isotonic calibration must be fit on a fixed held-out set; k-fold complicates this. Test isolation: with k-fold every sample eventually appears in a validation fold. Sample size: at 307K rows, variance on a single 20% val split is already low — k-fold adds computational cost without meaningfully reducing estimator variance.

Q — "Why TPE over random search?"

Random search samples uniformly with no memory of prior results. TPE builds a probabilistic model after each trial — it fits two density models over configurations that produced good/bad val AUC, proposes configurations where the good-to-bad ratio is highest. After even 3-4 trials, TPE starts exploiting productive regions. With a budget of 4-8 trials, sample efficiency matters.

Q — "val-vs-test PSI is 0.0002. Is that too good? Does something seem wrong?"

No — it is expected. Val and test are from the same stratified random split of the same dataset, drawn from the same distribution at the same point in time. PSI close to zero is the correct result when there is no temporal shift. The interesting PSI question would be comparing a new vintage to training — impossible here without timestamps. If val-vs-test PSI were high, that would suggest a bug in the split, not a real drift signal.

Q — "Are SHAP reason codes legally compliant adverse action notices?"

No. ECOA and Regulation B require adverse action notices with specific reason codes drawn from an approved list, reviewed by a licensed credit officer. SHAP-derived language is a draft aid — it helps a credit officer identify the right reason codes faster. Output is always tagged ASSISTIVE_ONLY and HUMAN_REVIEW_REQUIRED. A credit officer must review, confirm, and sign off before any applicant communication.

Q — "Why not C=1 in LogisticRegression for Platt?"

C is inverse regularisation in sklearn. C=1e6 means effectively zero regularisation — the sigmoid finds the best slope and intercept on the validation scores without penalty. We want the calibrator to fit the calibration data as closely as possible; the only guard against overfitting is the held-out test set evaluation (ECE_test=0.0034 confirms it generalises).

Q — "What is the difference between Brier score and ECE?"

Brier score is mean squared error of probability predictions — mean((p_i - y_i)^2) — a proper scoring rule penalising both miscalibration and poor discrimination. ECE is calibration-specific: bins predictions, computes the gap between mean predicted probability and observed DR per bin, averages across bins. A model can have low Brier but high ECE (good discrimination, bad calibration) or vice versa. We report both: Brier=0.0668 measures overall probabilistic accuracy; ECE=0.0034 confirms the probabilities are reliable.

19. Core ML Concept Drill

Q — "How does LightGBM differ from XGBoost architecturally?"

Two main differences. Tree growth: LightGBM uses leaf-wise (best-first) growth — splits the leaf producing the largest loss reduction regardless of depth. XGBoost uses level-wise growth — completes all splits at depth d before going to d+1. Leaf-wise converges faster but can overfit on small datasets. Split-finding: LightGBM uses histogram-based binning (256 bins), finding splits on bin boundaries. XGBoost exact split-finding evaluates every unique value. LightGBM also has GOSS (downsamples low-gradient instances) and EFB (bundles sparse features). Core gradient boosting math is the same.

Q — "What does scale_pos_weight do mathematically?"

scale_pos_weight multiplies the gradient and hessian of every positive-class (minority) sample by that factor before summing across the node. At scale_pos_weight=11.39 the model treats each defaulter as if it were 11.39 samples, making misclassifying a defaulter 11.39x more costly during training. It does not change the output probabilities directly — it changes what the trees optimise. Calibration is still needed afterwards.

Q — "What is a proper scoring rule? Is AUC one?"

A proper scoring rule is minimised by reporting the true conditional probability — it incentivises honest probability estimates. Brier score and log-loss are proper. AUC is NOT: you can maximise AUC with any monotone transformation of the true probabilities. AUC measures ranking accuracy, not calibration. This is why we do not select champion by AUC alone — a model with AUC=0.78 and ECE=0.29 is less useful than AUC=0.77 and ECE=0.003 for threshold-based policy decisions.

Q — "What is the bias-variance tradeoff in your specific context?"

With 140 features and 184K training rows, variance (overfitting) is the dominant risk. Train AUC=0.8526 vs val AUC=0.7734 — a 0.08 gap — is the variance signature. Managed with: monotone constraints (reduce effective complexity), Optuna searching over lambda/alpha/min_child_samples/subsample, and not over-tuning (4-8 trials produces a stable local optimum).

Q — "What is the difference between monotone constraints and feature selection?"

Feature selection removes features from the model entirely. Monotone constraints keep a feature but restrict the direction of its effect. A +1 constraint on BUREAU_OVERDUE_RATIO means the model can use that feature for any split, but only splits where higher overdue ratio increases predicted default probability are permitted. The model can capture non-linearities and interactions in the constrained direction — just cannot produce a split where higher overdue ratio leads to lower risk, which would be economically nonsensical.

20. Implementation Probes ("Did You Write This?")

Q — "What shape is the pred_contrib output for LightGBM?"

Shape (n_samples, n_features + 1). The last column is the bias term — the model base score (log-odds of the training prior). To get feature-level SHAP contributions, slice [:,:-1]. The sum of all columns including bias equals the raw log-odds prediction for that sample. Verify: booster.predict(X, raw_score=True) should equal contrib.sum(axis=1) for all samples.

Q — "What was the exact root cause of the pandas index error in SHAP?"

y_val was a pandas Series with original Home Credit row labels — integers like 289233, 95708, 302524 — because sliced from the full DataFrame without resetting the index. green_idx was a numpy array of 0-based positional integers from np.where. When you do y_val[green_idx], pandas interprets it as label-based lookup — it looks for rows labelled 0, 1, 2, 3. Those labels do not exist in a Series indexed at 289233. Fix: y_val.values converts to a plain numpy array; positional indexing works correctly.

Q — "What is the exact PSI formula you implemented?"

Ten equal-frequency bins computed from the reference distribution using np.percentile(ref, np.linspace(0, 100, 11)). Bin edges nudged (bins[0] -= 1e-9, bins[-1] += 1e-9) to ensure boundary values fall inside. np.histogram(ref, bins) and np.histogram(curr, bins) give counts; divide by n for proportions. Epsilon smoothing: replace any proportion below 1e-9 with 1e-9 to avoid log(0). PSI = sum over bins of (curr_pct - ref_pct) x log(curr_pct / ref_pct).

Q — "Walk me through what happens when Optuna runs a trial."

Optuna TPE sampler proposes a hyperparameter configuration. The objective function receives it, builds the model with those parameters, trains on the training set, evaluates on the validation set, returns the val AUC. Optuna logs the (hyperparams, val_AUC) pair. After ~5 trials, TPE fits two KDE models — over configurations that produced top-25% vs bottom val AUC — and proposes the next configuration by maximising the good-to-bad density ratio. study.best_params gives the best trial hyperparams. Direction is "maximize" since we optimise val AUC.

Q — "If you called booster.predict(X, pred_contrib=True) on a 2-class problem with 50 features, what are the dimensions?"

(n_samples, 51) — 50 feature contribution columns plus 1 bias column. For binary classification LightGBM returns a single output column per sample, not two — it is the log-odds of the positive class.

21. Behavioral / STAR Questions

Q — "Tell me about a bug you hit and how you diagnosed it."

The SHAP pandas index error is the best example. The symptom was a KeyError on integer indices — looks like a simple off-by-one but was a dtype mismatch. Three different approaches across separate sessions: list conversion, .iloc, explicit numpy casting. None fixed it because I was treating the symptom, not the cause. Diagnosis came from printing the actual index of y_val: index[:5] = [289233, 95708, 302524]. That is not a 0-based index — it is original Home Credit applicant IDs. green_idx contained 0-based positional integers. Pandas label lookup for label 1 in a Series indexed at 289233: not found. Fix was one line: y_val.values. Lesson: when debugging pandas indexing errors, print .index first.

Q — "What is the most governance-conscious decision you made?"

Selecting the champion by composite score rather than AUC alone. After tuning, LightGBM_monotonic val AUC=0.7734 and CatBoost=0.7708 — a gap of 0.0026. On AUC alone, barely a real difference. The composite includes calibration, KS, PR-AUC, and a governance premium for monotone constraints. By building the composite before running tuning, I committed to the selection criteria upfront — could not retroactively choose whatever metric made the preferred model win. That is the same discipline a real model risk committee expects.

Q — "What is the hardest limitation to defend?"

Reject inference. The model was trained on approved applicants only — Home Credit never observed outcomes for declined applicants. If the previous lending policy systematically declined a particular demographic, the model inherits that bias. There is no clean fix on this dataset. Semi-supervised reject inference methods exist but require strong assumptions about the declined population. I documented it as a material limitation in the model card. In any model trained on bank data, reject inference is a real problem that cannot be handwaved.

Q — "Tell me about a design decision you would change."

Using the external SHAP library at all for tree models. The library is valuable for model-agnostic explanations or visualisation tools. For LightGBM, pred_contrib is the native equivalent — same values, no dependency overhead. I planned to use the SHAP library because it is the standard portfolio recommendation, not because it added anything LightGBM could not do natively. Had I read the LightGBM pred_contrib documentation first, I would have used it from the start and avoided the index bug entirely.

22. Role Expansion: Fraud / MLOps / Risk Scoring

27A — Fraud Detection Roles

The bridge: PulseGuard solves imbalanced binary classification — 8.07% positive rate, scale_pos_weight=11.39, gradient boosting, Platt calibration, SHAP reason codes. Fraud detection is the same problem class at lower positive rate (0.1-5%). Methods are identical. Feature domain and regulatory vocabulary differ.

Question	Fraud-context answer
"Have you worked on fraud detection?"	Not directly — PulseGuard is credit default. Problem class is identical: rare positive label, gradient boosting, c
"How would you handle class imbalance in fraud?"	Same approach: scale_pos_weight. At 8% default rate I used 11.39. For 0.5% fraud rate: ~199. Platt/isotonic
"How would drift monitoring apply to fraud?"	PSI on score distribution plus top-k feature PSIs. Fraud pattern drift is faster — fraud rings adapt. Tighten PS

27B — MLOps / ML Platform Roles

The bridge: PulseGuard is a manually-implemented version of what an MLOps platform automates. Champion selection criteria = what Model Registry encodes. Evidence ledger = what MLflow experiment tracking stores. PSI/KS drift policy = what a monitoring dashboard surfaces.

Question	MLOps-context answer
"Walk me through model lifecycle experience."	End to end: data audit -> leakage-checked feature engineering -> baseline tournament -> validation-only Opt
"Have you used MLflow?"	Not MLflow specifically — PulseGuard evidence ledger is the manual equivalent. Every experiment result log
"What is a champion/challenger framework?"	Deploy a champion in production, route a fraction of traffic to a challenger. If challenger meets promotion crite

27C — Risk Scoring / Decision Science Roles

- Score-band thresholds set at PD=0.20 and PD=0.40 — interpretable meaning; business stakeholder can reason about "20% chance of default" without understanding the model
- Cost-sensitive decisioning: documented scenario analysis under C_bad=10, C_reject=1. Framework ready for real bank economics; inputs are scenario assumptions.
- Champion selection by composite score is decision science discipline: define criteria before running the experiment, not after

27D — Live Endpoint Deployment Caveat

Dimension	Live Endpoint	PulseGuard Champion
Model	G4 XGBoost + Platt	GP2 LightGBM_monotonic + Isotonic
Training data	synthetic_home_credit_like	Home Credit Default Risk
Rows	50,000	307,511
Features	28	140
Test AUC	0.6261	0.7769
Deployment status	LIVE at Cloud Run	NOT deployed (sklearn version incompatibility)

Root cause: GP2 pkl artifacts serialized with scikit-learn 1.7.2 (requires Python 3.10+). Deployment machine runs Python 3.9, which maxes out at sklearn 1.6.1. Cross-version pkl deserialization is not supported.

Fix if more time: Pin sklearn version at training time to match deployment. Or use ONNX export to decouple model artifact from sklearn version entirely.

23. Failures I'm Proud Of

Failure	Root Cause	Fix	Senior Behavior Demonstrated
DGP default rate overshoot (26% vs 8% target)	Logistic intercept too shallow (-2.8 instead of -4.2)	Binary search on intercept; N=200k for stability, 8 iterations	Verify data distributions before modeling
LightGBM early stopping bug (AUC=0.7203)	early_stopping_rounds fires at iteration 1 with scale_pos_weight	Remove early stopping; use n_estimators as Optuna parameter	Audit suspicious baselines before accepting them
SHAP pandas index bug (3 sessions)	y_val had non-sequential Home Credit IDs as index; green_idx had 0-based integers	y_val.values — drop index	Debug root causes, not symptoms; print .index first
Python 3.9 union type annotation crash	ColumnTransformer | None uses PEP 604 syntax (3.10+ only)	from __future__ import annotations (PEP 563)	Environment parity; know the Python version matrix
sklearn pkl version lock-out	Serialized with sklearn 1.7.2; deployed on Python 3.9/sklearn 1.6	Retrained P4 demo model on Python 3.9; documented version gap fully	Training/serving environment parity; version-pin artifacts
joblib vs pickle deserialization mismatch	app.py used pickle.load(); train_champion.py saved with joblib.dump()	joblib.load() throughout	Match your serializer; read the error traceback carefu

Portfolio-level discipline: None of these were hidden. Each one is documented in the gate logs, the claim boundary, or the failure archaeology document. A model that looks clean because failures were buried is more dangerous than one where failures are catalogued.

24. Gold Pass 5: LSTM Sequence Encoder Experiment

What Was Built

Component	Detail
Input data	installments_payments.csv — 13.6M rows, 339,587 applicants
Per-row features	days_late, payment_ratio, log_amt_instalment
Sequence length	Last 50 instalments per applicant, left-padded with zeros
Architecture	1-layer LSTM (input=3, hidden=64) -> Linear(64->32) -> tanh -> 32-dim embedding
Training	PyTorch, Colab T4 GPU, 15 epochs, BCEWithLogitsLoss(pos_weight=11.4), AdamW(lr=3e-3), gradient clipping (max
Integration	Embeddings appended to 140-feature LightGBM input -> 172 features; same GP2 Optuna hyperparameters with mor

Results

Metric	GP5 Challenger	GP2 Champion	Delta
Test AUC (calibrated)	0.7264	0.7769	-0.0505
Verdict	-	GP2 RETAINED	-

Why GP5 Did Not Win

- 35% of applicants have zero instalment history — zero-padded sequences encode noise, not signal, for 65,639 of 184,506 training applicants
- Scalar aggregates (INST_LATE_RATIO, INST_PAY_RATIO) are already in top-7 SHAP features in GP2 — LSTM learned a redundant representation
- Supervised LSTM on 8.1% imbalanced labels with 50-timestep sequences may not converge to embeddings more informative than handcrafted aggregates

What GP5 Adds to the Portfolio Story

- Ability to write PyTorch training loops with imbalance handling, early stopping, and gradient clipping
- Understanding of sequence preprocessing: padding, feature engineering at the row level
- Integration of a neural component with a tree ensemble pipeline
- Honest evaluation: the negative result is documented, not buried
- Concrete answer to "walk me through a neural component you built"

Key Q&A;

Q: "What would you do differently to make the LSTM work?"

Three things. First, restrict LSTM features to only the 65% of applicants who have instalment history — separate "no history" indicator for the others rather than zero-padding everyone. Second, try bidirectional LSTM or self-attention over the sequence (100+ payments window). Third, pre-train the encoder on a reconstruction task (autoencoder on payment sequences) before fine-tuning on the default TARGET — supervised training on 8% imbalance may converge to a less informative latent space.

25. Calibration Forensics: Isotonic Discovery

What Happened

The GP2 training pipeline evaluated both Platt (logistic regression) and isotonic regression calibration. The training report stated "Platt selected." The serving code was initially written using the Platt sigmoid formula. During Cloud Run deployment debugging, the serving probabilities diverged from the training calibrated probabilities by up to 0.53 at some score values — a massive serving gap.

Root Cause

Forensic inspection of champion_calibrated.pkl:

```
cal["selected"] # -> "isotonic" (NOT "platt")
```

The isotonic calibrator was actually selected at training time. The training report was wrong. The cal_probs in the pkl matched IsotonicRegression.predict() exactly (max diff = 0.0), and diverged from the Platt sigmoid (max diff = 0.53).

The Fix

```
iso_x = iso.X_thresholds_ # shape (134,) — raw probability thresholds iso_y = iso.y_thresholds_ # shape (134,) — calibrated probability values np.save("iso_x.npy", iso_x) np.save("iso_y.npy", iso_y) # At serve time — exact replication, zero sklearn: calibrated_prob = float(np.interp(raw_prob, iso_x, iso_y))
```

np.interp implements the same piecewise-linear interpolation that IsotonicRegression.predict() uses. Max difference between the two: 0.0 across all test values. This eliminates sklearn version lock at serve time entirely — calibration runs as a two-array numpy operation.

Interview Answer

Q: "You said Platt calibration in your writeup, but your code uses isotonic?"

> Yes — this is one of the honest failures I am glad I found. The training report said Platt was selected, but the pkl had cal["selected"] = "isotonic". When I deployed initial serving code using the Platt formula, calibrated probabilities diverged from training by up to 0.53 — a clear red flag. The forensic fix: read the actual artifact, not the report. Once I confirmed isotonic was the true calibrator, I extracted it as numpy threshold arrays and replicated it with np.interp. This eliminated the sklearn version dependency entirely. ECE on test is 0.0034.

25b. Implemented vs Future Roadmap

Component	Status	Safe Claim	Future Extension
Multi-table feature engineering	IMPLEMENTED	"140 features from 7 tables"	WOE/IV scorecard secondary model
Leakage audit	IMPLEMENTED	"10/10 PASS, safe_to_tune=true"	Re-audit after future feature additions
12-model baseline tournament	IMPLEMENTED	"12-model tournament, 2 hard-failures documented"	Neural baseline with GPU
Optuna HPO	IMPLEMENTED	"Validation-only Optuna TPE, 4-8 trials"	100-trial GPU search
Isotonic calibration	IMPLEMENTED	"ECE=0.0034 post-isotonic; numpy serving"	Beta calibration; group-specific calibration
Composite champion selection	IMPLEMENTED	"9-component composite score"	DeLong test significance
Score-band policy	IMPLEMENTED	"GREEN/AMBER/RED PD-semantic thresholds"	Threshold calibrated to real LGD
SHAP reason codes	IMPLEMENTED	"SHAP top-5 stable 30/30"	Full 61k-row SHAP computation
Fairness proxy audit	PARTIALLY VERIFIED	"Proxy audit; no amplification beyond DI"	Full DI analysis with protected-class labels
Drift/PSI baseline	PARTIALLY VERIFIED	"val-vs-test PSI=0.0002 STABLE"	Monthly monitoring dashboard; OOT validatio
RAG/LLM demo	SIMULATED	"6-case demo; abstain fires; ASSISTIVE_ONLY"	50+ doc corpus; live LLM integration
GP5 LSTM experiment	IMPLEMENTED	"LSTM challenger AUC=0.7264; GP2 retained"	Bidirectional LSTM; pre-trained encoder
Champion/challenger monitoring	FUTURE	-	Challenger promotion framework with DeLon
Production serving API	DEMO ONLY	"FastAPI on Cloud Run; G4 demo model AUC=0.6261"	Deploy champion after environment-parity fix
Reject inference	FUTURE	-	Semi-supervised correction
Real temporal validation	FUTURE	-	Requires timestamped dataset (HMDA candi
MLflow / model registry	FUTURE	-	Evidence ledger is the manual equivalent
Full fairness audit (DI ratio)	FUTURE	-	Requires protected-class demographic label

25c. Cloud Run API — Serving Architecture

Endpoint

https://pulseguard-api-98058433335.us-central1.run.app (revision 00006-zrm)

Model served: G4 XGBoost demo (AUC=0.6261, 28 features, 50k synthetic rows). **NOT** the GP2 LightGBM champion (blocked by sklearn version incompatibility — fully documented).

Request Format (POST /predict)

```
{ "AMT_CREDIT": 450000, "AMT_INCOME_TOTAL": 180000, "AMT_ANNUITY": 22500, "AMT_GOODS_PRICE":
400000, "DAYS_BIRTH": -14000, "DAYS_EMPLOYED": -2000, "EXT_SOURCE_1": 0.65, "EXT_SOURCE_2": 0.55,
"EXT_SOURCE_3": 0.70, ... (28 features total) }
```

Response Format

```
{ "applicant_id": "demo_001", "raw_score": 0.134, "calibrated_pd": 0.112, "score_band": "GREEN",
"governance_tag": "ASSISTIVE_ONLY", "human_review_required": false, "shap_top3": [ {"feature":
"EXT_SOURCE_2", "contribution": -0.41, "direction": "risk_reducing"}, {"feature":
"CREDIT_TO_ANNUITY", "contribution": 0.18, "direction": "risk_increasing"}, {"feature":
"DAYS_EMPLOYED", "contribution": -0.14, "direction": "risk_reducing"} ], "adverse_reason_draft":
null, "llm_memo": null, "model_version": "G4_XGB_demo_v1", "disclaimer": "NOT_FINAL_DECISION.
Subject to credit officer review." }
```

Architecture Components

Component	Implementation	Notes
Web framework	FastAPI + Uvicorn	app.py in /src
Containerisation	Docker (python:3.9-slim base)	Dockerfile at repo root
Deployment	Google Cloud Run (us-central1)	Revision 00006-zrm
Feature pipeline	src/feature_pipeline.py	ColumnTransformer; from __future__ import annotations for Python 3.9
Calibration	iso_x.npy + iso_y.npy; np.interp	Zero sklearn dependency at serve time
SHAP at serve	booster.predict(X, pred_contrib=True)	Top 3 by |SHAP| per request
Model artifact	outputs/models/g4_champion.pkl (joblib)	Loaded with joblib.load() — matches joblib.dump() at train time
Monitoring	Request logs in Cloud Logging	No automated PSI/drift; manual review cadence

Known Deployment Issues Fixed

<table>
<tr><th>Issue</th><th>Root Cause</th><th>Fix Applied</th></tr>
<tr><td>UnpicklingError: STACK_GLOBAL</td><td>app.py used pickle.load(); train used joblib.dump()</td><td>Switched to joblib.load() throughout</td></tr>
<tr><td>TypeError: unsupported | operand</td><td>ColumnTransformer | None is Python 3.10+ syntax</td><td>from __future__ import annotations (PEP 563)</td></tr>
<tr><td>Champion pkl unloadable</td><td>sklearn 1.7.2 pkl incompatible with Python 3.9</td><td>Retrained G4 demo model on Python 3.9; serving gap documented</td></tr>
</table>

25d. Resume Bullets — Ready to Use

Short Bullets (1-line each)

- Built end-to-end credit-risk governance pipeline (Home Credit, 307k applicants) — LightGBM + Isotonic — AUC=0.7769 — ECE=0.0034 on 61k holdout
- Engineered 140 features across 7 relational tables (bureau, instalment, POS, credit card); composite behavioural risk score; FOIR/LTV/DTI proxies
- Delivered SR 26-2-aligned governance stack: SHAP reason codes, fairness proxy audit, score-band policy, PSI drift baseline, local RAG/LLM governance assistant

Strong Bullets (2-3 lines)

Credit-risk ML governance stack, Home Credit Default Risk: Engineered 140 features from 7 relational tables (57.4M rows); ran 12-model baseline + Optuna-tuned 5-model tournament; champion LightGBM with monotone constraints achieves AUC=0.7769, ECE=0.0034 post-isotonic on 61k held-out test set; selected by 9-component composite including calibration, explainability, and adverse-reason readiness — not AUC alone.

End-to-end leakage-audited pipeline: Pre-tuning 10-check leakage audit (safe_to_tune=true) ensured TARGET exclusion, val-only calibration, and single test-set evaluation; 60/20/20 stratified split with zero entity overlap; test set evaluated exactly once; all results traceable to outputs/evidence/ artifacts.

SHAP governance layer with bootstrap stability: Computed reason codes via LightGBM pred_contrib for 4 local applicant cases; bootstrapped global feature importance across 30 resamples (500 samples each) — top-5 features appear in all 30 bootstraps; EXT_SOURCE_MEAN rank-1 in 30/30; adverse-action drafts tagged ASSISTIVE_ONLY, HUMAN_REVIEW_REQUIRED.

Calibration and score-band policy: Isotonic regression calibration reduces ECE from 0.296 (raw) to 0.0034 (test); served as numpy interp arrays — zero sklearn dependency at runtime; GREEN<0.20 (89.7% of test, DR=5.8%), AMBER 0.20-0.40 (9.8%, DR=27%), RED>=0.40 (0.5%, DR=54%).

Imbalanced-data modeling: Handled 8.07% default rate (scale_pos_weight=11.39); diagnosed LightGBM early-stopping bug with scale_pos_weight (fires at iteration=1); fixed by treating n_estimators as Optuna hyperparameter; AUC improved from 0.7203 (baseline, bug present) to 0.7734 (tuned, fix applied).

Local RAG/LLM governance assistant: BM25 policy retriever over 5-document corpus; abstain threshold (BM25<0.25) prevents hallucinated citations; 6-case demo (approve/review/decline/conflict/OOD/drift); LLM drafts ECOA-style adverse-action memos for credit officer review — ASSISTIVE_ONLY, HUMAN_REVIEW_REQUIRED enforced.

GP5 Neural Experiment: Built 1-layer LSTM encoder on 13.6M instalment payment rows (PyTorch, Colab T4 GPU); 32-dim embeddings appended to 140-feature LightGBM; challenger AUC=0.7264 vs baseline 0.7769 (delta=-0.0505); negative result documented — 35% zero-padding and redundant signal are root causes; GP2 champion retained.

Unsafe Bullets to Avoid

- "Built production credit scoring model" — not production
- "Model approved for lending" — not approved
- "AUC 0.85+ on credit data" — champion is 0.7769
- "100 Optuna trials completed" — actual: 4-8 per model
- "Fair model across protected classes" — proxy audit only
- "CatBoost champion at AUC=0.7716" — wrong champion and wrong metric (that is the untuned baseline CatBoost)
- "Live endpoint serves the GP2 champion" — endpoint serves G4 XGBoost demo; champion pkl is unloadable on Python 3.9

26. Final Defense Checklist

Item	Status
60-second opener ready (Section 17, Q1)	READY
2-minute walkthrough ready (Section 17, Q2 extended)	READY
Every metric memorised: AUC=0.7769, PR-AUC=0.2628, KS=0.4141, Brier=0.0668, ECE=0.0034	MEMORISED
Can explain champion selection (composite, not AUC; monotone governance premium)	READY
Can explain 57.4M rows correctly (raw data volume; aggregated to 307K modelling grain)	READY
Can explain limitations (reject inference, no temporal, proxy fairness, CPU trial count)	READY
Can explain why not production (portfolio project; no independent validation)	READY
Can defend LLM boundary (ASSISTIVE_ONLY; 6 cases; LLM never decides)	READY
Can defend fairness limits (no protected-class labels; proxy audit only)	READY
Can answer adversarial follow-ups (k-fold, stratify, PSI=0.0002, SHAP legality)	READY
Can answer core ML concepts (LightGBM vs XGBoost, TPE, bias-variance, proper scoring rules)	READY
Can answer implementation probes (pred_contrib shape, index bug, PSI formula)	READY
Can answer behavioral questions (bug story, design decision, champion selection rationale)	READY
Can explain GP5 LSTM experiment and why it failed (35% zero-padding, redundant signal)	READY
Can explain calibration forensics (isotonic vs Platt, numpy serving, max diff=0.0)	READY
Know forbidden claims and avoid them under pressure	CRITICAL

Resume-Safe One-Liner

"Built end-to-end credit-risk governance pipeline (Home Credit, 307k applicants, 57.4M rows) — LightGBM with monotone constraints + isotonic calibration; AUC=0.7769, ECE=0.0034 on 61k holdout; selected via 9-component composite score; SHAP reason codes stable 30/30 bootstraps; fairness proxy audit, PSI drift baseline, local RAG/LLM governance assistant (ASSISTIVE_ONLY); GOLD checkpoint at 89.3%."